

simplicio bench v13 — 5-side comparison (baseline / cli / cli+sp / cli+ag / cli+sp+ag)

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Sides

sem = baseline (raw goal). **com** = cli alone (6-layer). **sp** = sp runtime wrapping cli (composition). **ag** = cli + verify-loop (up to 3 attempts with PHPUnit/missing-pattern feedback). **spag** = sp-wrapped cli as the verify-loop seed (full stack composition + retry).

Models tested

- meta-llama/llama-3.2-3b-instruct (n=12 cases)
- google/gemma-3-4b-it (n=12 cases)
- qwen/qwen-2.5-coder-32b-instruct (n=12 cases)

Headline — pct per side, both metrics

Model	Metric	base	cli	cli+sp	cli+ag	cli+sp+ag	Δcli	Δsp	Δag	Δsp+ag
llama-3.2-3b-instruct	exec	8%	8%	8%	8%	8%	+0	+0	+0	+0
llama-3.2-3b-instruct	regex	32%	69%	67%	88%	76%	+37	+35	+56	+44
gemma-3-4b-it	exec	33%	66%	50%	66%	41%	+33	+17	+33	+8
gemma-3-4b-it	regex	42%	92%	88%	92%	92%	+50	+46	+50	+50
qwen-2.5-coder-32b-instruct	exec	8%	16%	16%	16%	16%	+8	+8	+8	+8
qwen-2.5-coder-32b-instruct	regex	34%	44%	38%	82%	80%	+10	+4	+48	+46

Verify-loop convergence (avg attempts)

Model	cli+ag avg	cli+sp+ag avg
llama-3.2-3b-instruct	2.83	2.83
gemma-3-4b-it	1.67	2.17
qwen-2.5-coder-32b-instruct	2.67	2.67

Cost & latency (exec, per call avg)

Model	Side	tokens/call	ms/call
llama-3.2-3b-instruct	baseline	1079	1757
llama-3.2-3b-instruct	cli alone	1273	1764
llama-3.2-3b-instruct	cli + sp	1976	822
llama-3.2-3b-instruct	cli + ag	3103	5224
llama-3.2-3b-instruct	cli + sp + ag	2828	2584
gemma-3-4b-it	baseline	1449	10010
gemma-3-4b-it	cli alone	1655	9575
gemma-3-4b-it	cli + sp	2655	11013
gemma-3-4b-it	cli + ag	3091	19180
gemma-3-4b-it	cli + sp + ag	4616	21105
qwen-2.5-coder-32b-instruct	baseline	1005	8580
qwen-2.5-coder-32b-instruct	cli alone	1215	8597
qwen-2.5-coder-32b-instruct	cli + sp	2268	8656
qwen-2.5-coder-32b-instruct	cli + ag	2463	25794
qwen-2.5-coder-32b-instruct	cli + sp + ag	3562	25986

Per-task × model (exec)

Format: **base/cli/cli+sp/cli+ag/cli+sp+ag** — P=pass, .=fail.

Task	llama-3.2-3b-instruct	gemma-3-4b-it	qwen-2.5-coder-32b-instruct
password_strength	./././././.	./P/P/P/.	./././././.
password_require_symbol	./././././.	./././././.	./././././.
env_get_int	./././././.	./P/P/P/P	./P/P/P/P
env_get_bool	./././././.	./P/./P/.	P/P/P/P/P
admin_only_allowed_roles	P/P/P/P/P	P/P/P/P/P	./././././.
rate_limit_bucket_key	./././././.	./P/P/P/P	./././././.
base_repository_build_where_sql	./././././.	./././././.	./././././.
router_has	./././././.	P/P/P/P/P	./././././.
bugfix_password_policy_lowercase	./././././.	P/P/P/P/P	./././././.
password_assess	./././././.	P/P/./P/.	./././././.
base_repository_build_update_sql	./././././.	./././././.	./././././.
router_extract_params	./././././.	./././././.	./././././.